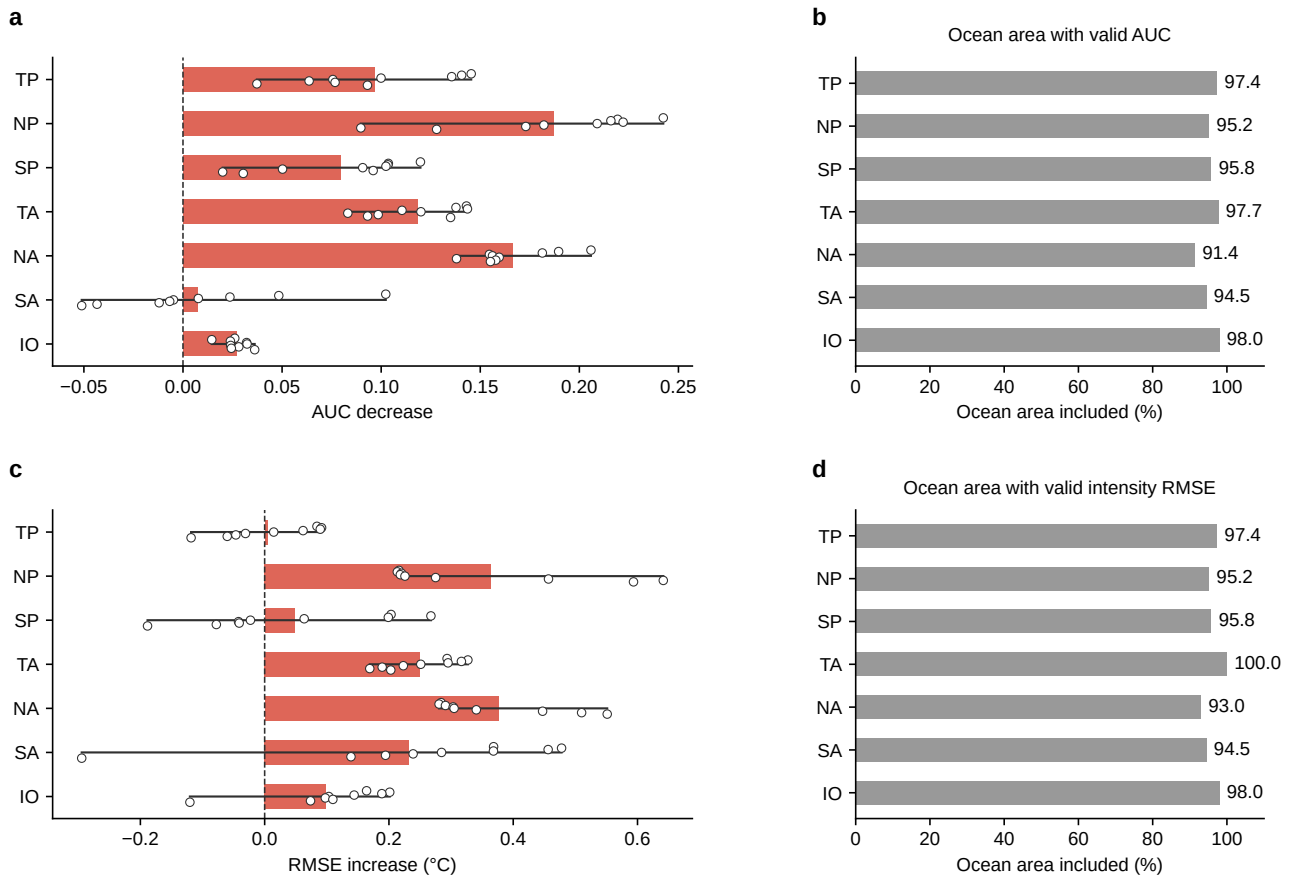
Supplementary Fig. 16 | Regional skill changes and valid ocean-area coverage



a, Mean 1997–1998/2015–2016 neighbourhood AUC minus 2023–2024 AUC. c, 2023–2024 conditional intensity RMSE minus the two-event reference. Bars show region summaries; small points and whiskers retain all nine lead values and their minimum–maximum range. Positive differences indicate poorer forecasts in 2023–2024; negative differences indicate improvement. b,d, Cosine-latitude-weighted area coverage where all three events and all nine leads are evaluable for the respective metric. Denominators are the full original analysis-region ocean areas, not the already valid cells. TP, tropical Pacific; NP, North Pacific; SP, South Pacific; TA, tropical Atlantic; NA, North Atlantic; SA, South Atlantic; IO, Indian Ocean. These ±20° sensitivity-analysis boundaries differ from the ±23.5° masks in main Figs. 3–4 and Supplementary Fig. 15.

All spatial comparisons use April–September of the following year and 11° × 11° same-basin ocean neighbourhoods. AUC requires both observed classes; missing comparisons stay undefined. Neighbourhoods overlap and months/leads are dependent. Regional AUC differences are not additive contributions to the global monthly spatial-AUC gap. Lead ranges are descriptive, not confidence intervals.